



Dwight Look College of
ENGINEERING
TEXAS A&M UNIVERSITY

ECEN 404

Status Update 2

Group #17: Patent Miner

List of team members:

Christian Casteel

CJ Vines

Kosi Ezewudo

Sponsor: Dr. Reddy Narasimha

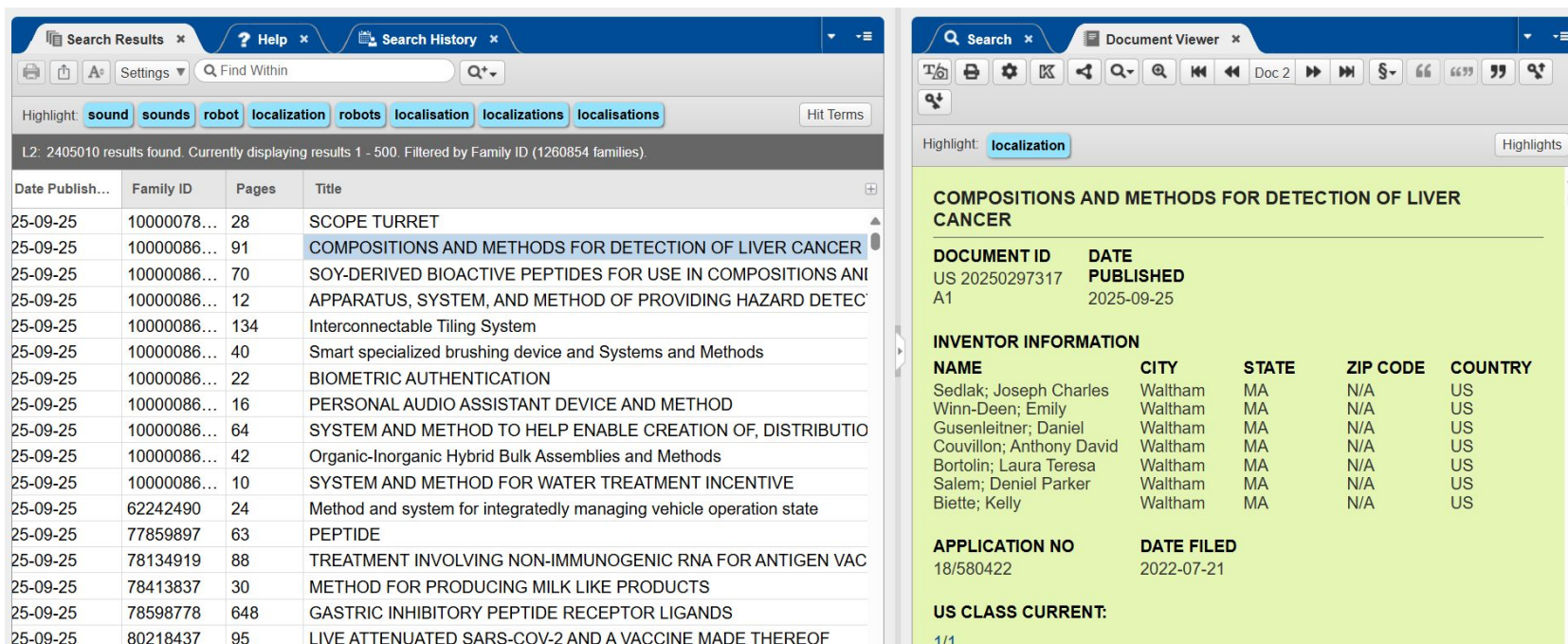
Project Summary

Problem statement:

- The current keyword search implemented by the United States Patent and Trademark Office (USPTO) fails to capture contextual meaning, leading to inaccuracy of search results and expensive billing rates for litigation professionals.

Solution:

- Development of a patent database mining tool that optimizes the patent search process by retrieving contextually relevant patents faster and more accurately compared to a traditional database.



The screenshot displays a patent search interface. The left pane shows search results for the keyword 'localization'. The right pane shows the detailed view of the selected patent, 'COMPOSITIONS AND METHODS FOR DETECTION OF LIVER CANCER'.

Search Results (Left Pane):

Date Publish...	Family ID	Pages	Title
25-09-25	10000078...	28	SCOPE TURRET
25-09-25	10000086...	91	COMPOSITIONS AND METHODS FOR DETECTION OF LIVER CANCER
25-09-25	10000086...	70	SOY-DERIVED BIOACTIVE PEPTIDES FOR USE IN COMPOSITIONS ANI
25-09-25	10000086...	12	APPARATUS, SYSTEM, AND METHOD OF PROVIDING HAZARD DETEC
25-09-25	10000086...	134	Interconnectable Tiling System
25-09-25	10000086...	40	Smart specialized brushing device and Systems and Methods
25-09-25	10000086...	22	BIOMETRIC AUTHENTICATION
25-09-25	10000086...	16	PERSONAL AUDIO ASSISTANT DEVICE AND METHOD
25-09-25	10000086...	64	SYSTEM AND METHOD TO HELP ENABLE CREATION OF, DISTRIBUTIO
25-09-25	10000086...	42	Organic-Inorganic Hybrid Bulk Assemblies and Methods
25-09-25	10000086...	10	SYSTEM AND METHOD FOR WATER TREATMENT INCENTIVE
25-09-25	62242490	24	Method and system for integratedly managing vehicle operation state
25-09-25	77859897	63	PEPTIDE
25-09-25	78134919	88	TREATMENT INVOLVING NON-IMMUNOGENIC RNA FOR ANTIGEN VAC
25-09-25	78413837	30	METHOD FOR PRODUCING MILK LIKE PRODUCTS
25-09-25	78598778	648	GASTRIC INHIBITORY PEPTIDE RECEPTOR LIGANDS
25-09-25	80218437	95	LIVE ATTENUATED SARS-COV-2 AND A VACCINE MADE THEREOF

Patent Details (Right Pane):

COMPOSITIONS AND METHODS FOR DETECTION OF LIVER CANCER

DOCUMENT ID US 20250297317
DATE PUBLISHED 2025-09-25

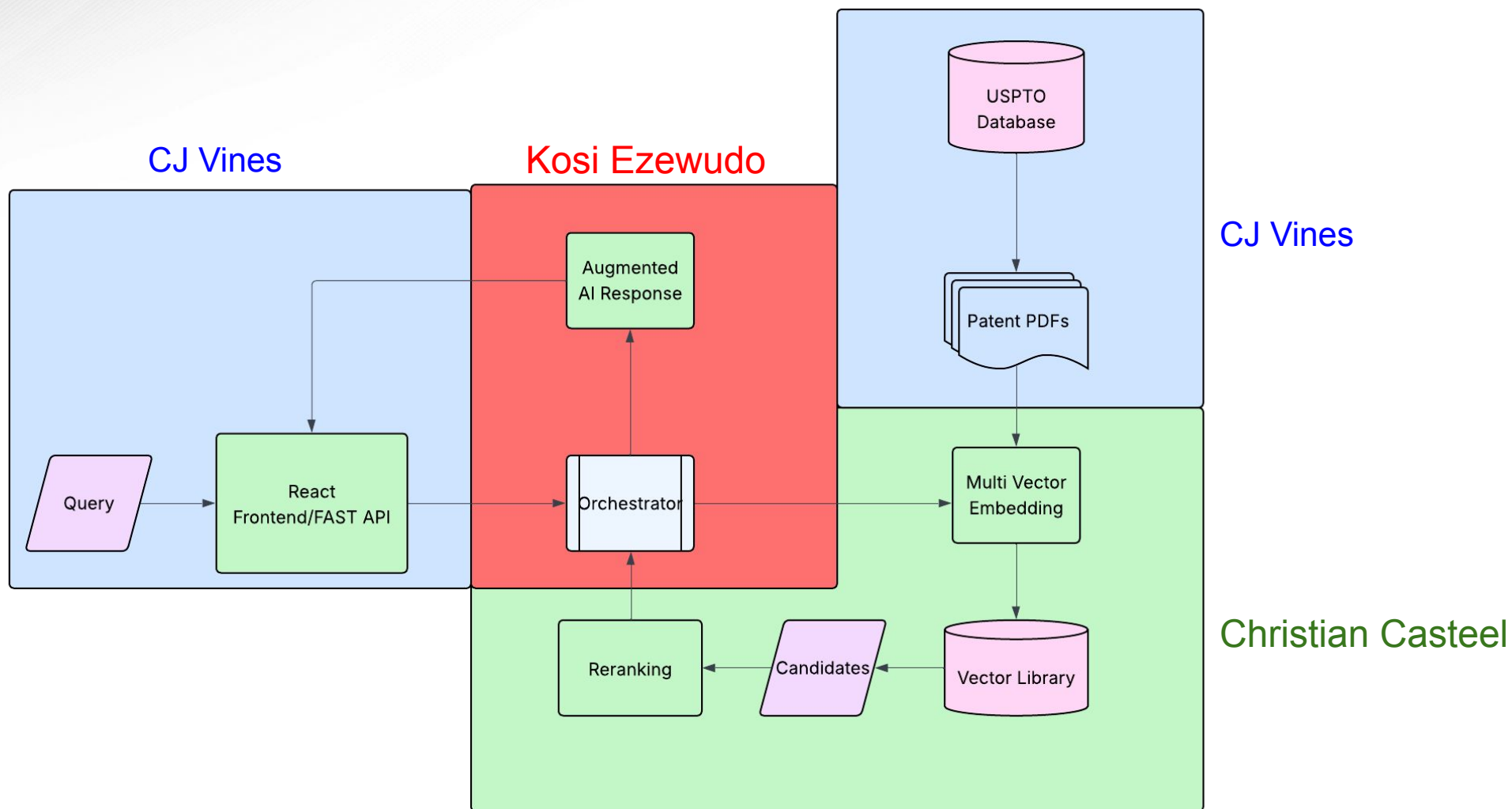
INVENTOR INFORMATION

NAME	CITY	STATE	ZIP CODE	COUNTRY
Sedlak; Joseph Charles	Waltham	MA	N/A	US
Winn-Deen; Emily	Waltham	MA	N/A	US
Gusenleitner; Daniel	Waltham	MA	N/A	US
Couvillon; Anthony David	Waltham	MA	N/A	US
Bortolin; Laura Teresa	Waltham	MA	N/A	US
Salem; Deniel Parker	Waltham	MA	N/A	US
Biette; Kelly	Waltham	MA	N/A	US

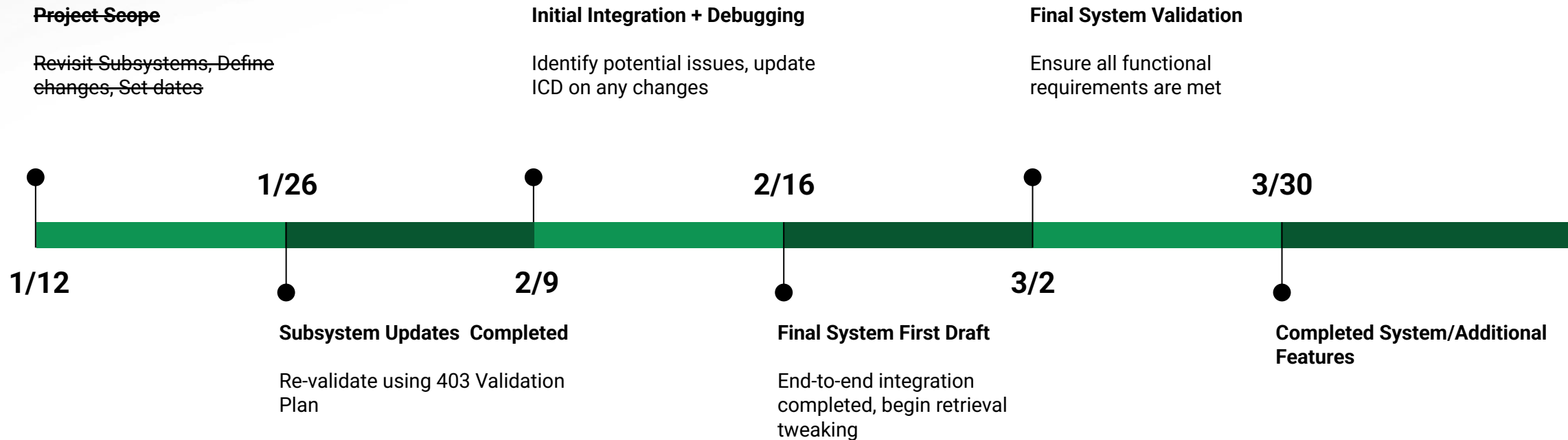
APPLICATION NO 18/580422
DATE FILED 2022-07-21

US CLASS CURRENT: 1/1

System Overview



Project Timeline



Embedding Pipeline and Storage

Christian Casteel

Accomplishments since last update	Ongoing progress/problems and plans until the next presentation
• Implemented reranking using CJ's LMDB database • Verified that storage was mean-pooled multi vectors, switched to MUVIRA dense multi vectors + LMDB multi vector database • Began validation of new embedding and retrieval pipeline	• Evaluate recall@100 and other metrics @10 to determine whether retrieval or reranking is a bottleneck • Find queries with the worst metrics, search for weaknesses • Finalize pipeline and complete validation

Embedding Pipeline and Storage

Christian Casteel

nDCG@10 heatmap across different set sizes

	Set Size			
Method	20 queries	40 queries	100 queries	nDCG@10
MUVERA only	0.357	0.301	0.125	0.5
CoBERT 128 f32	0.037	0.072	0.007	0.4
CoBERT 768 f32	0.439	0.413	0.148	0.3
CoBERT 768 f16	0.439	0.413	0.148	0.2
CoBERT 768 i8	0.439	0.414	0.148	0.1

Accuracy gain from reranking on 40 query set

	Metrics			
Method	P@10	R@10	nDCG@10	MRR@10
MUVERA	0.085	0.319	0.301	0.364
CoBERT 768 i8	0.11	0.413	0.414	0.479
Relative gain	29.40%	29.40%	37.20%	31.60%

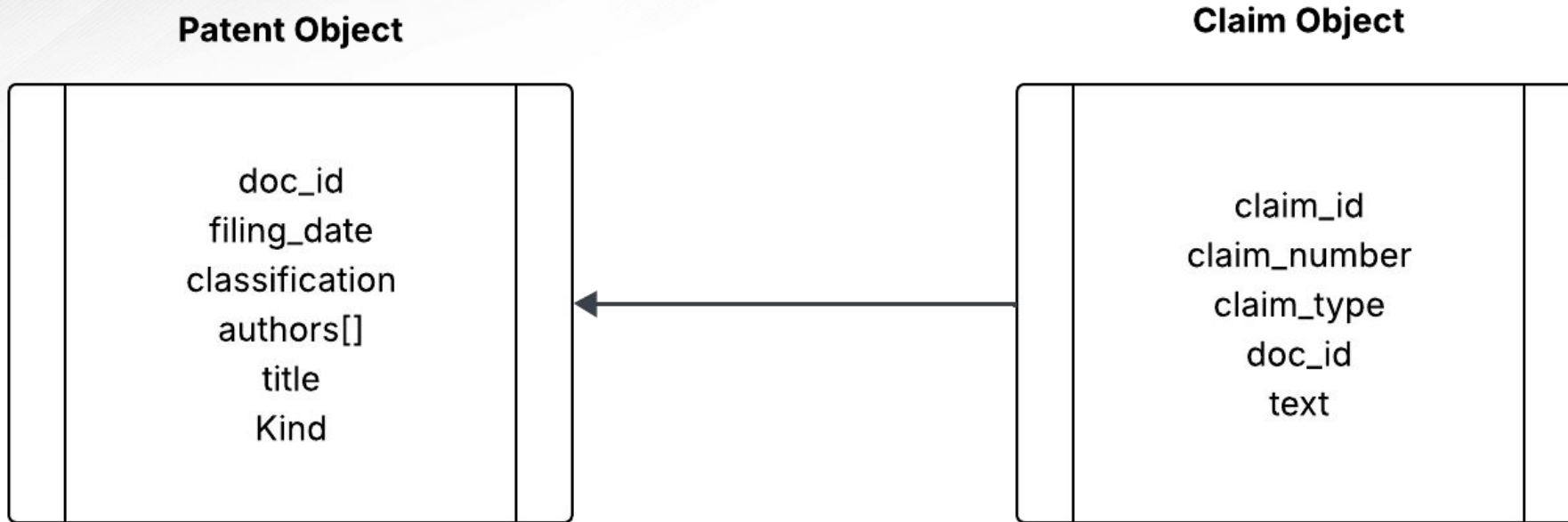
These results are preliminary, and are subject to change

Frontend/Preprocessing

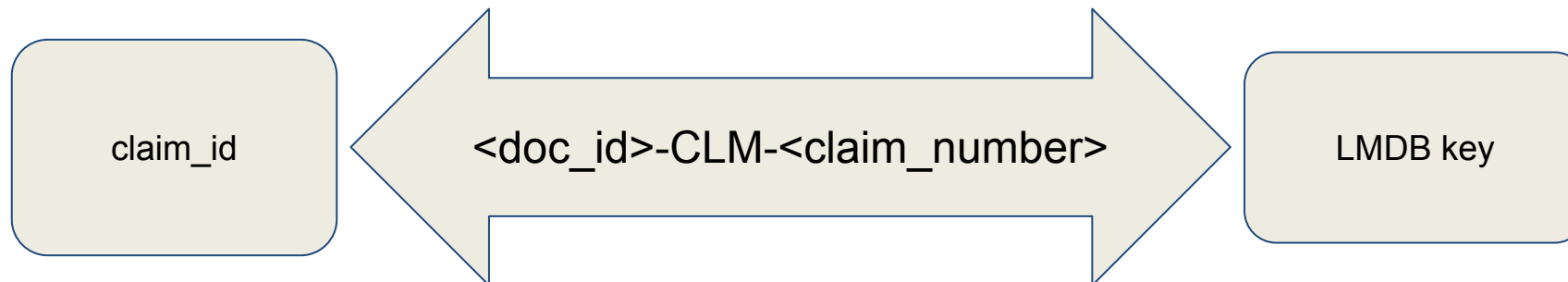
CJ Vines

Accomplishments since last update	Ongoing progress/problems and plans until the next presentation
• Redesigned Weviate database for more efficient storage of metadata • Set up CUDA wheel for streamlining embedding process • Implemented LMDB storage system for storage of CoLBERT vectors • Implemented Tailscale VPN to allow teammates access to Weaviate image and LMDB shards • Generated simple patent metrics to analyze scalability concerns • Semantic chunking implementation	• Orchestrator integration into API route • Scaled patent ingest to acceptable size • Assisting other subsystems in their implementation

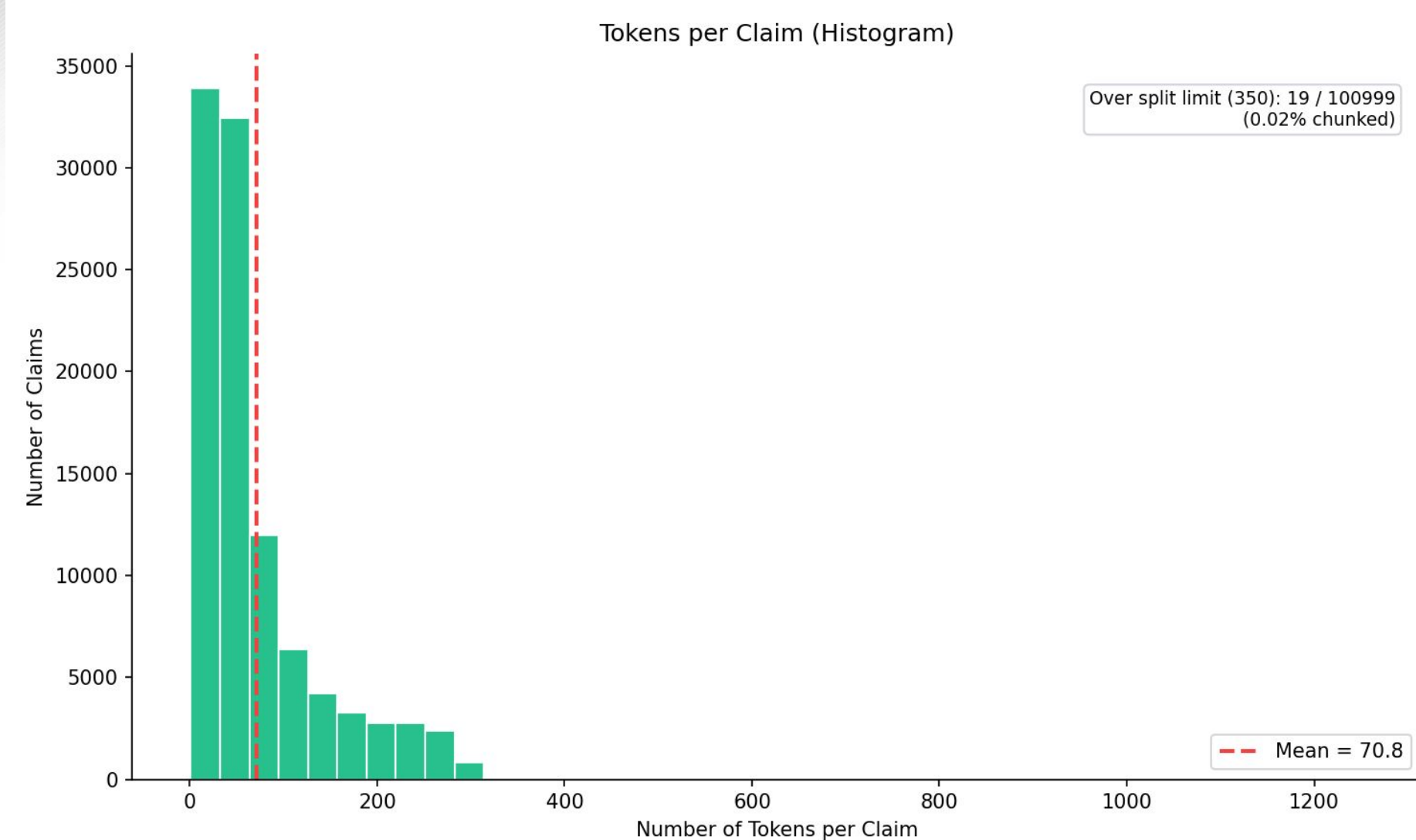
Weaviate+LMDB Organization



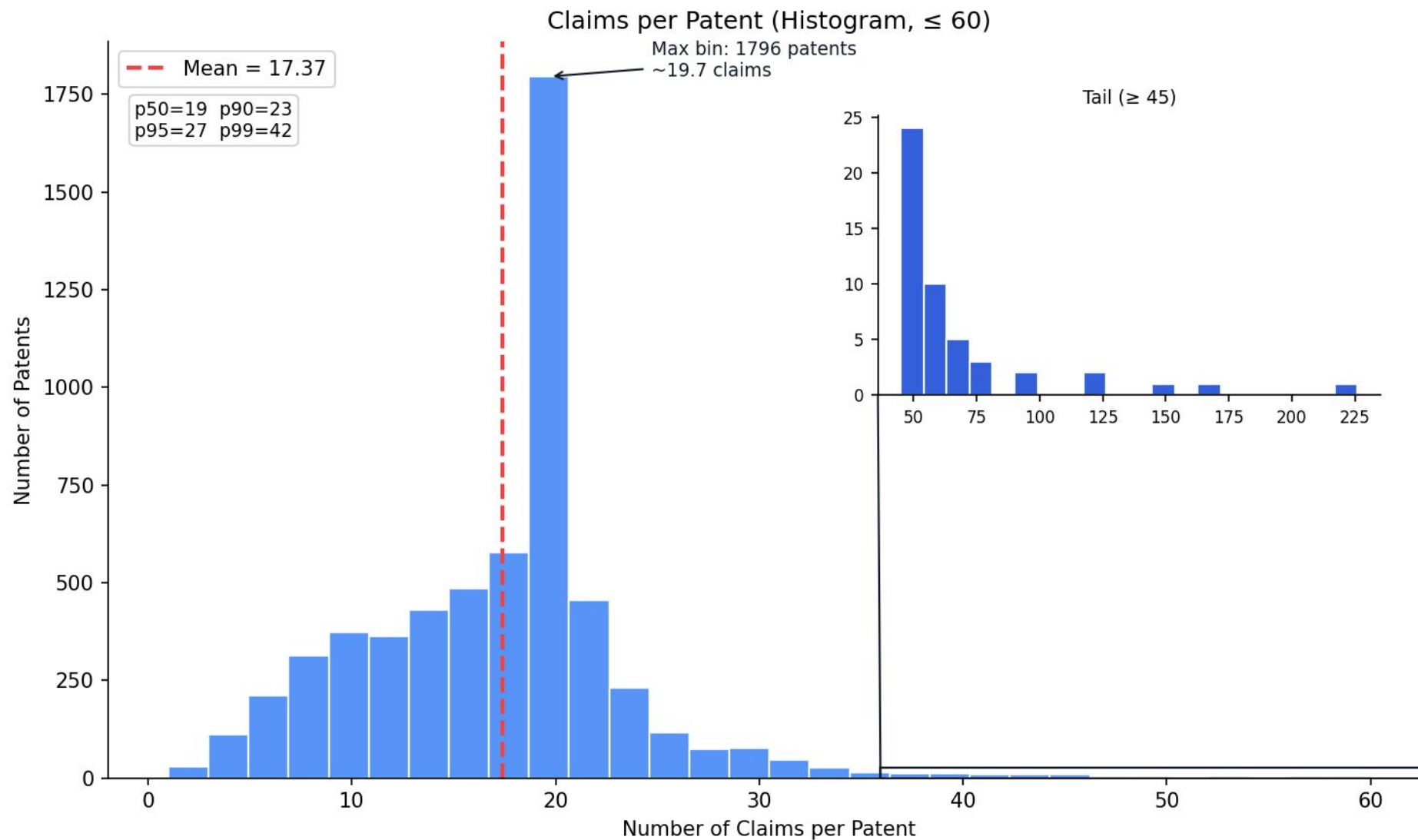
claim_id serialization for one-to-one relation between DBs



Interesting Stats, pt 1.



Interesting Stats, pt 2.



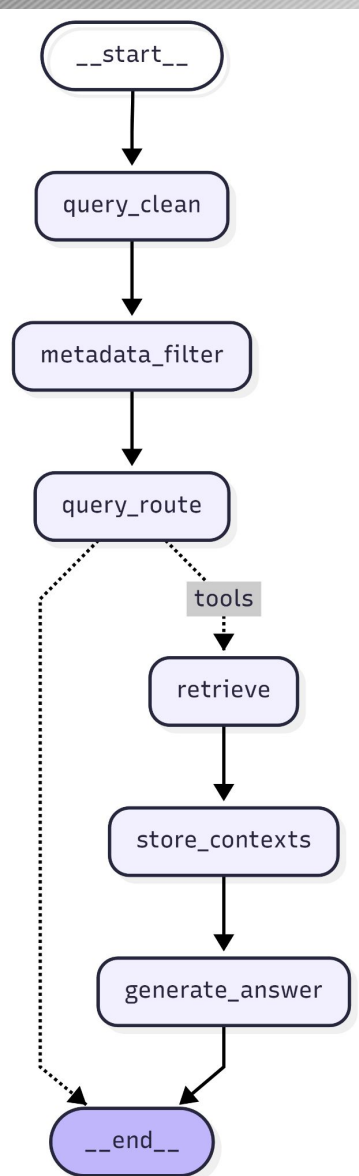
Retrieval Augmented Generation (RAG) Subsystem

Kosi Ezewudo

Accomplishments since last update	Ongoing progress/problems and plans until the next presentation
• Finalized and integrated Metadata extraction layer to RAG pipeline • Implemented RAGAS Evaluation pipeline for RAG system testing • Implemented an LLM-API request timing layer • Finalized subsystem and performed validation tests	• Refine full subsystem pipeline • Begin working on integration with other subsystems

Retrieval Augmented Generation Subsystem

Kosi Ezewudo



Metrics	Score
Faithfulness	0.8799
Answer Relevancy	0.7296
Metadata Extraction Accuracy	0.8512
Query Cleaning Accuracy	0.92

Metrics for RAG Subsystem

```

32 author_name="Carmine D'Alessandro" filing_date=None title='Methods for adjusting the phase of a carrier replica signal' document_id=None
33 author_name='Yuan Hse Chang' filing_date=None title='Clamp-on type pipeline measuring devices and fixing structures of the same' document_id=None
34 author_name='Summit Datta' filing_date='April 2024' title='electrical system' document_id=None
35 author_name='Kai Jiang and Yao Sun' filing_date=None title='Electronic seal label apparatus, electric motor assembly and electronic seal label verification me
36 author_name=None filing_date=None title='Adaptive lens assemblies including polarization-selective lens stacks for augmented reality display' document_id=None
37 author_name=None filing_date=None title='Predicting ceramic colloidal suspension stability for extrusion-based additive manufacturing' document_id=None
38 author_name='Sascha Klement' filing_date=None title='Method and system for determining a condition of a geographical line' document_id=None
39 author_name='Junya Onoue' filing_date=None title='Outboard motor' document_id=None
  
```

Sample Extracted Metadata from User Query

Execution Plan

[illegible]

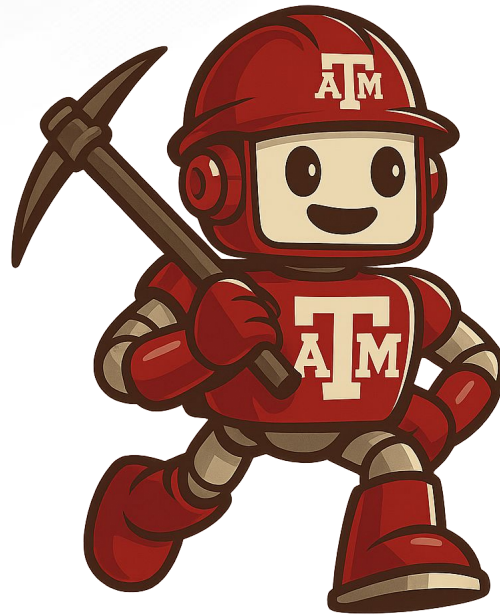
Validation Plan

Paragraph #	Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
3.2.1.2.	API/Pydantic Test	API gives correct response depending on the route taken 100% of the time	Send input query and data of different variations including edge cases like empty queries, queries with special characters, and get requests of the wrong shape	Tested	CJ Vines
3.2.1.1.	Determinism Test	Preprocessing Subsystem resulting chunks consistent across runs on the same dataset	Calculate SHA-256 hash of entire list of dictionaries after chunking and compare hash across runs to ensure download pipeline is deterministic. If any run is inconsistent, the test fails	Tested	CJ Vines
3.2.1.1.	Metadata Extraction	Stored metadata is verified to be at least 99% accurate	Compare metadata after chunking to metadata called from USPTO API, aiming for 99% accuracy	Tested	CJ Vines
3.2.1.4.	Full System Stress Test	Multiple simultaneous API calls are made successfully no data loss within 1ms	Make multiple API calls simultaneously, handle each concurrently in order.	Untested	CJ Vines & Kosi Ezewudo & Christian Casteel
3.2.1.5.	Embedding Reranking Test	"Golden" query - chunk pairs must have $\geq .05$ Recall@k, Mean Reciprocal Rank (MRR) $\geq .5$, and Normalized Discounted Cumulative Gain (NDCG) of .5	k = 10 target chunks at minimum with an equal number of matching queries. Calculate Recall@k, MRR, and NDCG for dataset. "Golden" set will include 50 hand selected patents.	Untested	Christian Casteel
3.2.1.5.	Embedding Accuracy Test	For a set of chunks, Hit@k of 1 and precision at k above .5. Relevance determined by patent id	Query containing information from multiple patents with similar CPC codes will test retrieval. Calculate Hit@k and Precision@k on top k chunks where k = 10.	Untested	Christian Casteel
3.2.1.1.	Failure Compliance Test	Pipeline can take in incorrect patent data and continue embedding without failure. Also will flag patents with false data	5 fake patents containing empty sections and incorrect CPC codes will be sent through the pipeline.	Untested	Christian Casteel
3.2.1.6.	Metadata Integrity Test	All stored metadata must be 100% consistent with metadata listed on a patent	Select 50 patents, specifically patents with generally unique CPCs and numbers of authors, then compare all metadata fields to the same patents on the USPTO database	Untested	Christian Casteel
3.2.1.1.	Download and Embedding Pipeline Stress Test	A USPTO bulk dataset is downloaded, embedded, and stored without failure	A USPTO bulk dataset download is conducted via fronetend download script. The bulk dataset should contain minimum 1000 patents.	Untested	Christian Casteel & CJ Vines
3.2.1.6.	Query Metadata Extraction	Extract all relevant metadata from user query, and use the metadata to filter patent search	Execute patent search using queries containing various metadata including prior date, patent ID, CPC etc.	Untested	Kosi Ezewudo
3.2.1.3.	Query Cleaning	Removes typos, sentence fragments etc. from user query	Execute patent search using grammatically incorrect queries	Untested	Kosi Ezewudo
3.2.1.4.	Generation Accuracy and Quality Test	Generated responses must have $\geq .8$ Answer relevancy scores, $\geq .8$ Diversity and Novelty	Query Patent Miner using golden set of patents. Generate responses and calculate Answer relevancy, Diversity and Novelty scores	Untested	Kosi Ezewudo & Christian Casteel & CJ Vines



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY



Thank you!